THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue ____________________
Seat Number __________
Student Number |__|__|__|__|__|__|__|__|
Family Name ____________________
First Name ____________________

School of Civil Engineering
Semester Two Examinations, 2024
CIVL6415 Traffic Analysis and Simulation

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•Specified written or printed material permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

One A4 sheet of handwritten or typed notes double sided is permitted
Unannotated bilingual dictionary

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x Gradescope Bubble Sheet
1 x Gradescope Answer Booklet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer Multiple Choice Questions 1 - 20 in the Gradescope Bubble Sheet provided.

Answer Problem 1 - 4 in the Gradescope answer booklet provided. Write answers in the space provided for each question. If you need additional space, use the extra pages at the end of the answer booklet, clearly indicating the question number you are answering.

For Examiner Use Only

Question	Mark

Total _________

Part A: Multiple Choice Questions. To be completed in the Gradescope Bubble Sheet.
(Part A is worth 50/100 for the final exam. Each of the multiple choice questions is worth 2.5 marks.)

1) A traffic stream on a freeway has an average spacing of 50m and a flow of 1000 veh/h. What is the space mean speed for this traffic stream?
   a) 30 km/h
   b) 35 km/h
   c) 40 km/h
   d) 45 km/h
   e) 50 km/h

2) A section of highway has the following flow-speed relationship, where density (k) is in veh/km and flow (q) is in veh/h.

$$q = 300v - 10v^2$$

   What is the maximum speed?
   a) 30 km/h
   b) 35 km/h
   c) 40 km/h
   d) 45 km/h
   e) 50 km/h

3) Which one of the statements below about the fundamental traffic flow variables is TRUE?
   a) Flow is the product of density and time mean speed.
   b) Space mean speed is the arithmetic mean of spot speeds over a certain period of time.
   c) Space mean speed is the harmonic mean of vehicle speeds at a given instant of time.
   d) Time mean speed is always greater than space mean speed.
   e) The conversion from occupancy to density requires average vehicle length and detector length.

4) Three cars are travelling on a 5 km circular track at constant speeds of 90 km/h, 120 km/h and 150 km/h, respectively. What is the time-mean speed?
   a) 90 km/h
   b) 105 km/h
   c) 120 km/h
   d) 135 km/h
   e) Not enough information has been provided.

5) Which statement below about dynamic traffic models is TRUE?
   a) In dynamic traffic models, travel times are fixed regardless of traffic flow changes.
   b) Dynamic traffic models allow for variations in travel times based on traffic conditions.
   c) Dynamic traffic models assume that all traffic links have the same traffic conditions at a given instant of time.
   d) Dynamic traffic model require one single path between origin and destination points.
   e) Dynamic traffic models do not consider variations in traffic demand over time.

6) Which statement below about traffic models is TRUE?
   a) Macroscopic models are static models developed mainly for planning purposes.
   b) Microscopic models assume vehicles move at the same speed while travelling on the same link.
   c) Microscopic models update the acceleration of each vehicle considering the local conditions surrounding each vehicle.
   d) Mesoscopic models adopt a continuous fluid approach for vehicle flow modelling.
   e) Mesoscopic models represent position, speed and acceleration of individual vehicles.

7) Which statement below about traffic models is TRUE?
   a) Microscopic models build on continuous fluid approximation.
   b) Macroscopic models build on aggregated representation of speed and user variables.
   c) Mesoscopic models allow the vehicles to make lane changes in roadway sections.
   d) Microscopic models require discrete representation of space.
   e) Mesoscopic models build on continuous fluid approximation.

8) Which statement below about transport system analysis is TRUE?
   a) Transport system analysis focuses only on short-term traffic patterns and ignores long-term planning.
   b) Effective transport system analysis includes evaluating the impact of land use patterns and demand fluctuations over time.
   c) The interaction between transport infrastructure and travel demand is ignored in most system analyses.
   d) Transport system analysis determines congestion levels based only on public transport options.
   e) None of the above.

9) Which statement below about IDM car following models is NOT true?
   a) IDM considers human factors through the desired speed and the desired space headway.
   b) IDM is a very popular car following model and used as the default car-following model in SUMO.
   c) The desired spacing in IDM is a function, rather than a constant value.
   d) When the preceding vehicle is far away, the distance-related term in the IDM equation becomes negligible and the model performs as a free flow model where the desired speed of the driver governs the acceleration.
   e) When accelerating on a free road from a standstill, the vehicle described by IDM starts with the desired acceleration.

10) Which statement below about car following is NOT true?
   a) Two primary goals of car following models are safety and efficiency of individual vehicles' movement.
   b) The basic unit of car following models is the driver-vehicle.
   c) Rules governing car following are typically simplified as a set of mathematical models that aim to reproduce the longitudinal and lateral interactions between a pair of vehicles.
   d) Most of the existing car following models are continuous-time car following models.
   e) Generally, discrete-time car following models are less accurate than continuous-time car following models; however, Gipps' car following model is discrete-time, and has been widely used.

11) Which statement below about GM car following models is NOT true?
   a) As one of the oldest car-following models, GM car following models are formulated in the stimulus-response framework.
   b) GM5 is a generic car following structure such that GM1-GM4 can be regarded as special cases.
   c) According to GM4, two vehicles can get arbitrarily close as long as they are travelling at the same speed.
   d) GM4 is a continuous-time car-following model and adopted in the traffic simulation package VISSIM as the default car-following model.
   e) There exist inherent linkages between GM models and macroscopic traffic flow models.

12) Which statement below about Gipps' car following models is NOT true?
   a) Gipps' model is the most popular safety distance model by hypothesizing that the driver reacts to spacing relative to the preceding vehicle, rather than to the relative speed, and that car-following behaviour can be described using a single function, regardless of traffic condition.
   b) Gipps' model assumes that during the reaction time, the subject vehicle will continue its previous kinematic dynamics (e.g., acceleration or deceleration).
   c) Gipps' model assumes that there is a delay for the subject vehicle to execute acceleration or deceleration, and that during this delay, the subject vehicle cruises at a constant speed.
   d) The safe speed estimated by Gipps' model can generally guarantee that a minimum gap between the subject vehicle and the preceding vehicle can be kept even in the worst case scenario.
   e) Gipps car following model is locally stable.

13) Traffic on a three-lane freeway observes a triangular flow-density relationship with free-flow speed v_f = 100 km/h, capacity c = 2000 veh/h/lane, and jam density k_j = 80 veh/km/lane. What is the density corresponding to congested traffic flow of 3000 veh/h?
   a) 120 veh/km
   b) 150 veh/km
   c) 180 veh/km
   d) 210 veh/km
   e) 240 veh/km

14) Arrival traffic flow is 1000 veh/h, from time t = 0 to time t =1hr. After t =1hr, vehicles arrive at a (lesser) rate of 600 veh/h. Considering a capacity of 800 veh/h, what is the maximum delay? Hint: Consider A-D curves.
   a) 0.05 h
   b) 0.10 h
   c) 0.15 h
   d) 0.20 h
   e) 0.25 h

15) Which one of the statements below about macroscopic models is TRUE?
   a) First-order models capture smooth changes of trajectories along shockwaves.
   b) First-order models allow vehicles to make decisions with respect to traffic conditions ahead of them.
   c) First-order models may produce capacity drop.
   d) Second-order models may produce non-stationary traffic states.
   e) Second-order models are called second-order, because they consider hierarchy of vehicle classes.

16) Which one of the statements below about ramp metering is TRUE?
   a) Ramp metering is not useful, because they do not affect outflow from merge areas.
   b) Ramp metering applications have limited success, because they cannot increase the flow in upstream off-ramps.
   c) Ramp metering algorithms allow vehicles to enter freeway faster.
   d) Isolated (single) ramp metering applications are criticised because they usually transfer congestion from freeways to urban areas.
   e) Coordinated ramp metering algorithms are designed to effectively use the storage capacity in coordination with arterial traffic signals.

17) Which one of the statements below about equilibrium conditions is TRUE? Assume there is demand only between one O-D pair and there are multiple alternative paths.
   a) In user equilibrium conditions, more than one path might be used.
   b) In user equilibrium conditions, the travel times on the used routes might vary.
   c) In system optimum conditions, the travel times on the used routes must be the same.
   d) In system optimum conditions, no user has an incentive to change their route choice.
   e) In user equilibrium conditions, all alternative paths must be used.

18) Which one of the statements below about traffic assignment is NOT true?
   a) Braess paradox states that an addition to the capacity will always improve traffic under system optimum conditions.
   b) Braess paradox implies that total travel time will always increase with every addition of capacity.
   c) The fixed-point solution refers to a solution that does not significantly change with an additional iteration of the assignment procedure.
   d) Traffic assignment addresses the interaction between departure time choice and travel times in the network.
   e) Dynamic traffic assignment models are often easier to solve than their static counterparts due to the dynamic modelling of traffic characteristics.

19) Which one of the statements below about traffic assignment in Aimsun is TRUE?
   a) Aimsun uses only fixed route choices that do not change as vehicles progress through the network.
   b) Aimsun can consider real-time instantaneous travel times in the network when making route choices.
   c) Aimsun allows vehicles to make route choices only when they enter the network.
   d) Aimsun’s simulation capabilities are limited to urban traffic networks and do not support freeway modelling.
   e) None of the above.

20) Which one of the statements below about traffic assignment is TRUE?
   a) In a dynamic traffic assignment model, vehicles are routed based on time-varying demand and traffic conditions.
   b) Traffic assignment models always assume that each vehicle travels independently without considering interactions with other vehicles.
   c) Dynamic traffic assignment models assume that traffic conditions remain constant throughout the modelling period, regardless of time or demand changes.
   d) In a stochastic traffic assignment model, all trips are assigned to the shortest path available, ignoring variations in travel time.
   e) Traffic assignment concerns the selection of destinations in the network.

Part B: Short Answer Questions

Part B is worth 50/100 for the final exam. Marks are indicated next to each question. Please answer the Gradescope answer booklet provided. Write answers in the space provided for each question. If you need additional space, use the extra pages at the end of the answer booklet, clearly indicating the question number you are answering.

Problem 1

An observer standing at a point along a three-lane roadway collects spot speed data for all vehicles as they cross the observation point for 15 minutes. All vehicles in lane 1 are travelling at 60 km/h, all vehicles in lane 2 are travelling at 80 km/h, and all vehicles in lane 3 are travelling at 100 km/h. The headway between vehicles is 6 sec/veh across all lanes.

1) What is the density? **[2 marks]**
2) What is the flow? **[2 marks]**
3) What is the time mean speed? **[3 marks]**
4) What is the space mean speed? **[3 marks]**

[10 marks]

Problem 2

Consider two vehicles travelling on a road. Vehicle *i* is following vehicle *(i-1)*. The acceleration of vehicle *i* is described by the car-following model below:

$$a_i(t) = \frac{\beta\,[v_0 - v_i(t)][v_{i-1}(t) - v_i(t)]}{\alpha\,[x_{i-1}(t) - x_i(t)]}$$

Where $a_i(t)$ is the acceleration of vehicle *i* at time *t*, $v_i(t)$ and $v_{i-1}(t)$ are the speeds of vehicle *i* and *(i-1)* at time *t*, $x_i(t)$ and $x_{i-1}(t)$ are the positions of vehicle *i* and *i-1* at time *t*, α and β are two parameters ($\alpha = 0.5\,s$ and $\beta = 0.4\,m.$), and v_0 is the desired speed of vehicle *i*. The initial conditions are: the initial locations $x_{i-1}(t=0) = 10\,m$, $x_i(t=0) = 0\,m$ and the initial speeds $v_{i-1}(t=0) = 20\,m/s$, $v_i(t=0) = 25\,m/s$. Suppose $v_0 = 33\,m/s$.

1) In this car-following model, what are the stimuli, and how does each of these stimuli affect the estimated acceleration? **[5 marks]**

2) Determine the acceleration vehicle *i* will adopt at *t=0* if this vehicle's behaviour can be perfectly described by this car-following model. **[5 marks]**

3) If you use this model to perform a one-step simulation at *t=0* with a time step of 1 second, what will be the speed of vehicle *i* at *t=1* second, and what will be the distance travelled by vehicle *i* from *t=0* to *t=1* second? Assume vehicle *i* travels at the constant acceleration within each time step. **[5 marks]**

[15 marks]

Problem 3

Traffic on a two-lane freeway observes a triangular flow-density relationship with free-flow speed v_f=100 km/h, capacity c=2000 veh/h/lane, jam density k_j=80 veh/km/lane. Flow in the freeway from upstream is 3000 veh/h. At 8am, an accident occurs at point A. The vehicles involved close one lane of traffic for 30min and afterwards the scene is clear.

1) Sketch the space-time diagram with shockwaves (trajectories not needed). **[5 marks]**
2) Calculate the number of vehicles in the congested state at 8:45am. **[5 marks]**
3) Plot the trajectory of the first vehicle that arrives at the accident site after 8.30am without encountering any congestion. Calculate the time at which this vehicle crosses the accident location. **[5 marks]**

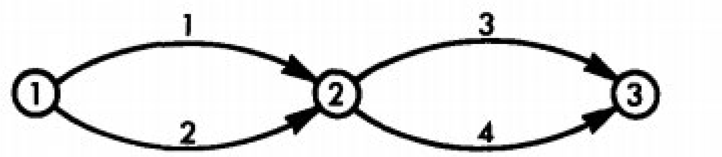
[15 marks]

Problem 4

Find the user equilibrium flow and travel times for the network shown in the figure below, where the link travel functions are

$$t_1 = 2 + x_1^2$$
$$t_2 = 3 + 2x_2$$
$$t_3 = 1 + 2x_3^2$$
$$t_4 = 2 + 4x_4$$

where x_1, x_2, x_3 and x_4 are expressed in thousands of vehicles per hour. The demand between nodes 1 and 3 is 4000 veh/h.
Hint: Assume all paths are used.



[10 marks]

END OF EXAMINATION